

TITLE: Use of Sedation and Analgesia in the Adult Intensive Care Unit

PROTOCOL: Sedation and Analgesia

PURPOSE: There are numerous different medications that can be used for sedation and analgesia in patients in the intensive care unit. The purposes of this protocol are to evaluate the various pharmacological options for appropriateness given certain patient co-morbidities and to review the indications and contraindications for the various pharmacological options.

APPROVED:



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DEFINITIONS:

A. RASS: Richmond Agitation-Sedation Scale

**Richmond Agitation-Sedation Scale (RASS)†**

|                                 |                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------|
| Combative (score of 4)          | Overtly combative, violent, immediate danger to staff                                              |
| Very agitated (score of 3)      | Pulls or removes tubes or catheters; aggressive                                                    |
| Agitated (score of 2)           | Frequent nonpurposeful movement, fights ventilator                                                 |
| Restless (score of 1)           | Anxious but movements not aggressive or vigorous                                                   |
| Alert and calm (score of 0)     | Alert and calm                                                                                     |
| Drowsy (score of -1)            | Not fully alert but has sustained awakening (eye opening or eye contact) to voice ( $\geq 10$ sec) |
| Light sedation (score of -2)    | Briefly awakens with eye contact to voice (<10 sec)                                                |
| Moderate sedation (score of -3) | Movement or eye opening to voice but no eye contact                                                |
| Deep sedation (score of -4)     | No response to voice but movement or eye opening to physical stimulation                           |
| Cannot be aroused (score of -5) | No response to voice or physical stimulation                                                       |

- B. PADIS
- C. BPS
- D. CPOT
- E. SAT
- F. SBT

**ANALGESIA:**

- 2018 PADIS guidelines: ABCDEF bundle
  - **A**ssess, Prevent and Manage Pain
    - Assessing for pain:
      - Behavioral Pain Scale (BPS):
        - 3 domains, 0-4 scoring in each (higher score->in more pain)
        - Facial expression, upper limb movement, compliance with mechanical ventilation
      - Critical Care Pain Observation Tool (CPOT):
        - 3 domains, 0-2 scoring in each
        - Facial expression, body movements, muscle tension
    - Utilizing **Analgo**sedation (analgesia FIRST or analgesia-only strategies to ensure proper treatment of pain prior to sedation)
      - Start with bolus dosing before continuous infusions
  - SAT/**S**BTs
  - **C**hoice of Analgesia and Sedation
    - IV opioids

|                                  | <b>Fentanyl</b>                                                                                                                                   | <b>Hydromorphone</b>                                                            | <b>Morphine</b>                                                                                                                                         |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dose</b>                      | Bolus: 25-50 mcg q30min<br>Infusion: 25-200 mcg/hr                                                                                                | Bolus: 0.2-1mg q2-3hr<br>Infusion: 0.5-2mg/hr                                   | Bolus: 2-4mg q1hr<br>Infusion: 1-5mg/hr                                                                                                                 |
| <b>Onset of action</b>           | 1-2min                                                                                                                                            | 5-10min                                                                         | 5-10min                                                                                                                                                 |
| <b>Duration of action</b>        | 30-60min (context sensitive half-time)                                                                                                            | 3-4hours                                                                        | 2-4hours                                                                                                                                                |
| <b>Pharmacokinetics</b>          | Hepatic (high 1st pass clearance (75%); reduced in CHF, liver disease)                                                                            | Hepatic via glucuronidation                                                     | - Hepatic metabolism via glucuronidation to M3G and M6G<br>- Metabolites are excreted via kidney and so accumulate in ARF                               |
| <b>Additional considerations</b> | - Chest wall rigidity/myoclonus (with rapid pushes at high dose)<br>- depressed respiratory outlasts the analgesic effect<br>- serotonin syndrome | - Histamine release<br>- less likely: hypotension, itching, urticaria, flushing | - Histamine release (more likely)<br>- M3G accumulation: neurotoxicity & myoclonus<br>- M6G accumulation: significant sedation & respiratory depression |

▪ PO opioids

|                        | <b>Oxycodone</b> | <b>Hydromorphone</b> | <b>Fentanyl patch (mg-mcg/hr)</b>                             | <b>MS-Contin</b> | <b>Methadone</b> | <b>Buprenorphine</b> |
|------------------------|------------------|----------------------|---------------------------------------------------------------|------------------|------------------|----------------------|
| <b>Dose</b>            | 5-15mg q4-6h     | 2-4mg q4-6h          | 0.3 (12),<br>0.6 (25),<br>1.2 (50),<br>1.8 (75),<br>2.4 (100) | 5-200 mg q12h    | 10-40mg q8h      | 4-24mg QD buccally   |
| <b>Onset of action</b> | 10-30min         | 30min                | 12-24 hours                                                   | 1-2 hour         | 2 hours          | 30-60min             |

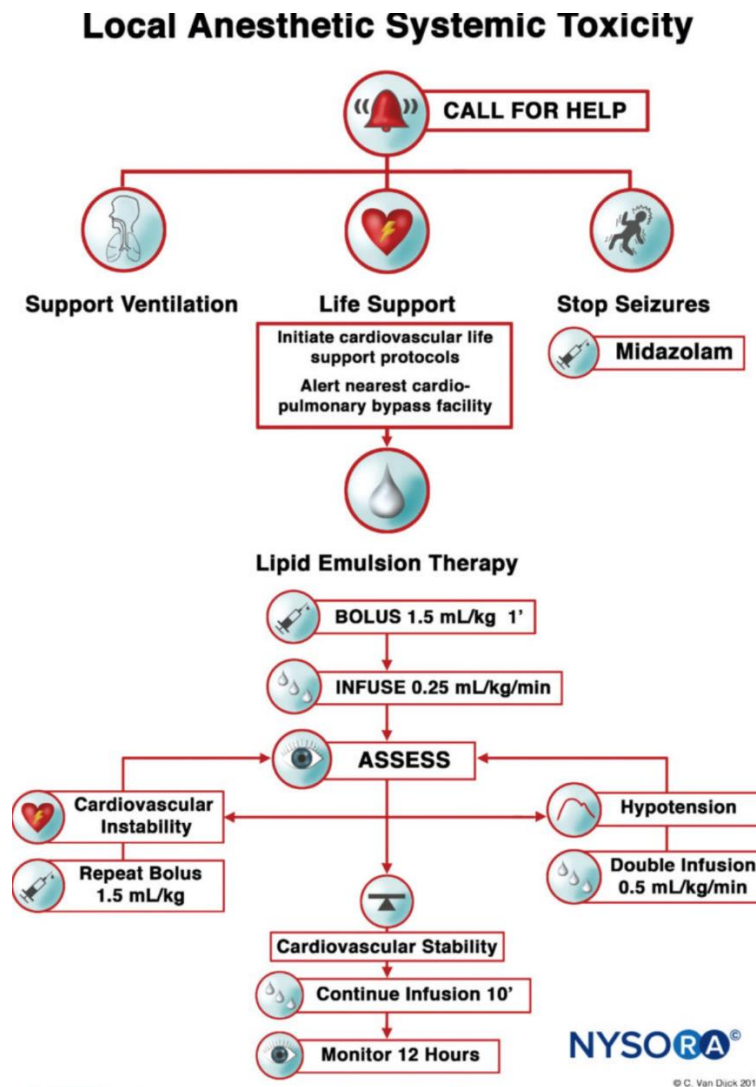
|                                  |                                                                                                       |                                            |                             |                                          |                                                                                                                                                            |                                                                                                                                                                                          |
|----------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------|-----------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Duration of action</b>        | 3-6 hours                                                                                             | 3-4 hours                                  | 3 days                      | 8-12 hours                               | 13-100 hours (variable)                                                                                                                                    | Low dose: 6-12 hours<br>High dose: 24-72 hours                                                                                                                                           |
| <b>Pharmacokinetics</b>          | - primarily by hepatic metabolism (various CYPs)                                                      | - primarily by hepatic metabolism into H3G | CYP3A4                      | (same as morphine)                       | CYP3A4 (does not need renal dosing)                                                                                                                        | CYP3A4                                                                                                                                                                                   |
| <b>Additional considerations</b> | - Start at ½ dosing in liver pts<br>- Unaffected by food<br>- 10% of the population need higher doses | - H3G accumulates in CKD                   | - Mostly used in cancer pts | - Risk of OD and death is 2x in ER vs IR | - Black box warning: QTc prolongation<br>- Same abuse potential as morphine<br>- Additional MOA: NMDA receptor antagonist, NE and 5-HT reuptake inhibition | - QTc prolongation, but much less than methadone<br>- Can precipitate withdrawal if on other opioids<br>- lowers seizures threshold<br>- reduced respiratory depression, GI side effects |

▪ Non-opioid Infusions

|                           | <b>Lidocaine</b>                           | <b>Ketamine</b>                                   | <b>Dexmedetomidine</b> |
|---------------------------|--------------------------------------------|---------------------------------------------------|------------------------|
| <b>Dose</b>               | Bolus: 1.5 mg/kg<br>Infusion: 1-2 mg/kg/hr | Bolus: 0.3-.5 mg/kg<br>Infusion: 0.1-0.2 mg/kg/hr | (As above)             |
| <b>Onset of action</b>    | 1-5 min                                    | 10-30 sec                                         |                        |
| <b>Duration of action</b> | 10-20min                                   | 5-15min                                           |                        |
| <b>Pharmacokinetics</b>   | 80% CYP3A4                                 | CYP3A4, CYP2B6                                    |                        |

|                                  |                                                                                                                                                                                          |                     |  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--|
| <b>Additional Considerations</b> | - Contraindications: heart blocks, CHF, liver impairment, severe electrolyte disturbances, acidosis<br>- Okay in renal patients for <24hr<br>- Local anesthetic systemic toxicity (LAST) | - CI: CAD, psych hx |  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--|

- LAST (local anesthetic systemic toxicity) algorithm<sup>11</sup>



- PO non-opioids

|                                      | <b>Acetaminophen</b>                        | <b>NSAID<br/>(Ketorolac)</b>                | <b>Gabapentin</b>                                                              | <b>Pregabalin</b>                                                              |
|--------------------------------------|---------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Dose</b>                          | 650-1000 mg q4-6h<br>(3-4g max) daily       | 15-30 mg q6h                                | 300mg TID starting<br>(3,600 max daily<br>dose)                                | 5-200 mg BID                                                                   |
| <b>Onset of action</b>               | 15 min                                      | 30min                                       | 2-3 hours                                                                      | 2-4 hours                                                                      |
| <b>Duration of action</b>            | 4-6 hours                                   | 4-6 hours                                   | 5-7 hours                                                                      | 8-12 hours                                                                     |
| <b>Pharmacokinetics</b>              | - 90% CYP2E1<br>- NAPQ1 toxic<br>metabolite | CYP2C8/9                                    | - NOT metabolized<br>in the body and<br>excreted<br>unchanged by the<br>kidney | - <2% metabolized<br>in liver, primarily<br>excreted<br>unchanged by<br>kidney |
| <b>Additional<br/>Considerations</b> | 2g max daily for liver<br>disease           | Avoid in cardiac,<br>renal, bleeding<br>pts | Renally dose<br>based on GFR,<br>even ESRD<br>patients can take                | Side effects:<br>dizziness,<br>peripheral edema                                |

▪ Muscle Relaxants

|                                      | <b>Cyclobenzaprine</b>                      | <b>Baclofen</b>                                          | <b>Methocarbamol</b>                                             | <b>Tizanidine</b>              | <b>Diazepam</b>                                            |
|--------------------------------------|---------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------|--------------------------------|------------------------------------------------------------|
| <b>Dose</b>                          | 5-10 mg q8h                                 | 5-20 mg<br>q8h                                           | 500-1500 mg q6-<br>8h                                            | 2-6 mg q6-<br>8h               | 1-10 mg q8h                                                |
| <b>Onset of action</b>               | 30-60min                                    | 1 hour                                                   | 30-60min                                                         | 1-2 hours                      | 15-60min                                                   |
| <b>Duration of action</b>            | 4-6 hours                                   | 2-6 hours                                                | 4-6 hours                                                        | 3-6 hours                      | 4-6 hours                                                  |
| <b>Pharmacokinetics</b>              | CYP3A4/CYP1A2                               | 15% liver<br>metabolism<br>~85%<br>excreted<br>unchanged | Hepatic<br>metabolism                                            | CYP1A2                         | CYP3A4 and<br>CYP2C19                                      |
| <b>Additional<br/>considerations</b> | - Beers List<br>- avoid in liver<br>disease | - Beers List<br>- GABA<br>agonist                        | - Beers List<br>- potentiated<br>CNS depression<br>when combined | - can<br>decrease<br>HR and BP | - Beers List<br>- Initial<br>metabolites<br>with very long |

|  |                                  |                     |                  |                                                   |                     |
|--|----------------------------------|---------------------|------------------|---------------------------------------------------|---------------------|
|  | - anti- cholinergic side effects | - severe coma in OD | with benzos/EtOH | - can have rebound hyper spasticity with stopping | half lives (>80hrs) |
|--|----------------------------------|---------------------|------------------|---------------------------------------------------|---------------------|

- Assess, Prevent and Manage **Delirium**
  - **E**arly Mobility and Exercise
  - **F**amily Engagement
- **Acute Pain Service (APS)** guidelines regarding anticoagulation and nerve blocks/catheters
    - All anticoagulation plans should be discussed with APS fellow and/or attending, as recommendations may vary on case-by-case basis
    - Recommend making patient NPO if planning for nerve block +/- catheter due to possibility of needing mild sedation
    - Currently, holding anticoagulation for **peripheral** nerve blocks and catheters is not recommended
      - Femoral, sciatic, popliteal sciatic, interscalene, infraclavicular
    - Holding anticoagulation for **neuraxial catheters and deep nerve blocks** is recommended as per the guidelines for common anticoagulants below<sup>10</sup>:

| Medication | Dose | Hold time BEFORE placement | Hold time AFTER placement | Hold time BEFORE | Hold time AFTER | Notes |
|------------|------|----------------------------|---------------------------|------------------|-----------------|-------|
|------------|------|----------------------------|---------------------------|------------------|-----------------|-------|

|                    |                   |                             |                                         | catheter<br>removal    | catheter<br>removal      |                                       |
|--------------------|-------------------|-----------------------------|-----------------------------------------|------------------------|--------------------------|---------------------------------------|
| <b>Heparin SQ</b>  | 5000u TID         | 6 hours                     | immediately                             | 4-6 hours              | immediately              | Check PLT if any<br>concern for HIT   |
| <b>Heparin IV</b>  | (infusion)        | 4 hours<br>AND check<br>PTT | Do not<br>restart<br>(case-by-<br>case) | n/a (case-<br>by-case) | 1 hour                   |                                       |
| <b>Enoxaparin</b>  | <0.5mg/kg/<br>day | 12 hours                    | immediately                             | 12 hours               | 12 hours                 |                                       |
| <b>Enoxaparin</b>  | >0.5mg/kg/<br>day | 24 hours                    | 12 hours                                |                        | 24 hours                 |                                       |
| <b>Clopidogrel</b> | (any)             | 7 days                      | Do not<br>restart                       | n/a                    | 6 hours for<br>75mg dose | 24 hours for<br>loading dose<br>>75mg |
| <b>Warfarin</b>    | (any)             | 5 days AND<br>check INR     | Do not<br>restart                       | n/a                    | immediately              |                                       |
| <b>Apixaban</b>    | (any)             | 72 hours                    | Do not<br>restart                       | n/a                    | 6 hours                  |                                       |
| <b>Rivaroxaban</b> | (any)             | 72 hours                    | Do not<br>restart                       | n/a                    | 6 hours                  |                                       |
| <b>Aspirin</b>     | 81mg of<br>325mg  | n/a                         | n/a                                     | n/a                    | n/a                      |                                       |
| <b>NSAIDs</b>      | any               | n/a                         | n/a                                     | n/a                    | n/a                      |                                       |

## SEDATION:

1. Monitoring sedation
  - a. Nursing: document Richmond Agitation and Sedation Scale (RASS) score **every 1 hour** for ICU level of care patients.
2. Medication options and their mechanism of action



|                                 | <b>Propofol</b>                                                                                               | <b>Dexmedetomidine</b>                                                                   | <b>Midazolam</b>                                                       | <b>Ketamine</b>                                                                                                                                             |
|---------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mechanism of Action</b>      | Gamma-aminobutyric acid (GABA) type A receptor agonist                                                        | Alpha-2 receptor agonist                                                                 | GABA type A receptor agonist                                           | N-methyl-D-aspartate (NMDA) receptor antagonist                                                                                                             |
| <b>Cardiac effects</b>          | - Vasodilation<br>- Bradyarrhythmias                                                                          | - Bradycardia<br>- Hypertension<br>- Hypotension                                         | - May cause hypotension                                                | - increased BP, HR, and CO (inc. O2 demand)<br>- Direct negative inotropic effects may predominate in those with pre-existing HF or catecholamine depletion |
| <b>Respiratory effects</b>      | Suppresses respiratory drive                                                                                  | No respiratory suppression                                                               | Suppresses respiratory drive                                           | Rarely causes respiratory depression (risk with rapid IV administration)                                                                                    |
| <b>Metabolism and excretion</b> | Hepatic metabolism (no dose adjustment), renal excretion                                                      | Hepatic metabolism (dose reduction recommended in severe insufficiency), renal excretion | Hepatic metabolism (clearance prolonged in cirrhosis), renal excretion | Hepatic metabolism (no dose adjustment), renal excretion                                                                                                    |
| <b>Additional information</b>   | - Highly lipophilic<br>- Risk of propofol infusion syndrome at high/prolonged doses<br>- hypertriglyceridemia | - Tachyphylaxis with prolonged use<br>- Rebound tachycardia                              | - Anterograde amnesia                                                  | - Emergence delirium<br>- Pro-sialagogue<br>- Potent bronchodilator<br>- analgesic properties (see below)                                                   |

### 3. Doses of sedative medications

#### a. Continuous infusions

- i. Notify physician if RASS goal not achieved after reaching maximum dose or if additional care parameters are outside of prescribed limits **(no change)**
- ii. Deviation from described titration parameters allowed as directed by the physician and must be documented in progress notes. **(no change)**

| Propofol ( <b>no change</b> )                                                                                                                                                                                                                                                                                                                                                    | Dexmedetomidine                                                                                                                                                                                                                                               | Midazolam ( <b>no change</b> )                                                                                                                                                                                                                                 | Ketamine (sedation dosing)                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>STARTING DOSE: based on weight range below or if started in the OR, maintain postoperative infusion rate</p> <ul style="list-style-type: none"> <li>&lt;120kg: 20mcg/kg/min</li> <li>121-150kg: 15mcg/kg/min</li> <li>&gt;150kg: 5mcg/kg/min</li> </ul> <p>TITRATION DOSE/FREQUENCY: 10mcg/kg/min every 5 minutes</p> <p>GOAL: RASS -1 to 0</p> <p>MAX DOSE: 50mcg/kg/min</p> | <p>STARTING DOSE: 0.5mcg/kg/hr for RASS 3+ or higher; or 0.3mcg/kg/hr for RASS 1+ to 2+</p> <p>TITRATION DOSE/FREQUENCY: 0.1mcg/kg/hr every 15 minutes</p> <p>GOAL: RASS 0 to -1</p> <p>MAX DOSE: 1.5mcg/kg/hr (1.8mcg/kg/hr with CCM attending approval)</p> | <p>STARTING DOSE: 2mg/hr or maintain postoperative infusion rate if started in OR</p> <p>TITRATION DOSE/FREQUENCY: 1mg/hr every 30 minutes</p> <p>GOAL: RASS -1 to 0</p> <p>MAX DOSE: 8mg/hr (up to 12mg/hr with CCM attending approval for ECMO patients)</p> | <p>STARTING DOSE: 0.3mg/kg/hr</p> <p>TITRATION DOSE/FREQUENCY: 0.1mg/kg/hr every 10 minutes</p> <p>GOAL: RASS 0 to -1</p> <p>MAX DOSE: 2mg/kg/hr</p> |

b. Intermittent boluses

- a. Propofol boluses should only be used if already on a continuous infusion (10-20mg every 1 hour PRN for RASS > +3, start with lowest dose and can give additional higher dose within range if RASS score is not reduced after 5 minutes.
- b. Bolus dose dexmedetomidine generally not recommended secondary to hemodynamic compromise concerns (bradycardia, hypertension, and hypotension)<sup>4</sup>
  - i. Can consider up to 1mcg/kg given over 10 minutes<sup>4</sup>

**c.** Maximum combined dose of midazolam boluses = 8mg/hr (**no change**)

| RASS score | Propofol          | Midazolam | Ketamine |
|------------|-------------------|-----------|----------|
| -5 to -2   | (not recommended) |           |          |

|    |                                   |                 |                                                                                                      |
|----|-----------------------------------|-----------------|------------------------------------------------------------------------------------------------------|
| -1 | None                              | None            | None                                                                                                 |
| 0  |                                   |                 |                                                                                                      |
| 1  | 10-20mg from infusion<br>q1hr PRN | 0.5mg           | Initial bolus: 0.5-2mg/kg<br>over 30-60 seconds.<br><br>Repeat boluses = 0.5-<br>1mg/kg <sup>5</sup> |
| 2  |                                   | 1mg (no change) |                                                                                                      |
| 3  |                                   | 2mg (no change) |                                                                                                      |
| 4  |                                   | 4mg (no change) |                                                                                                      |

#### 4. Sedation holiday

- a. Discontinue sedation every morning as per unit-specified practice and designated timeframe to perform neurological assessment or weaning parameters
- b. Resume sedation infusion or intermittent dose at one-half the previous dose following awakening<sup>1</sup>
- c. Sedation holiday protocol does not apply for patients with one of the following<sup>1,8</sup>
  - i. Severe agitation (RASS +3 or +4)
  - ii. Traumatic brain injury located in 4W
  - iii. Elevated intracranial pressure
  - iv. Status epilepticus
  - v. Requiring aggressive ventilation strategies for refractory hypoxemia (ECMO, bilevel, etc)
- d. Therapeutic or induced hypothermia
  - i. Use of neuromuscular blockade or barbiturate coma

KEYWORDS:

Sedation, agitation, delirium, pain control, acute/chronic pain consultation

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